MAT 201 - MATHEMATICAL METHODS I

LECTURE 5: TAYLOR SERIES

LECTURE C24-1-5 :: MON., JAN. 13, 2025 8:00 AM (WAT) - FRI., JAN. 17, 2025 6:00 PM (WAT)

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LESSON OUTLINE

- Overview of Taylor Series
- Taylor Series in One Dimension
- Taylor Series in Two Dimensions
- Summary

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OVERVIEW OF TAYLOR SERIES

- **Introduction**
  - Taylor series is a mathematical concept that plays a crucial role in approximating functions as an infinite sum of terms.
  - Named after the mathematician Brook Taylor, it provides a powerful tool for understanding and analyzing functions in calculus.

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OVERVIEW OF TAYLOR SERIES

- **Importance in Mathematics and Physics**
  - Function Approximation: Taylor series allow us to represent a wide variety of functions as infinite series, providing a useful way to approximate complex functions.
  - Analysis of Functions: By examining the terms of the series, we gain insights into the behavior of functions at specific points and their local properties.
  - Physics Applications: In physics, Taylor series are fundamental in approximating solutions to differential equations, aiding in the understanding of physical phenomena and modeling real-world problems.

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OVERVIEW OF TAYLOR SERIES

- **Motivation**
  - Understanding Local Behavior:
    - Taylor series enables us to understand the local behavior of a function near a specific point.
    - Provides a powerful tool for analyzing functions in the vicinity of a chosen point.

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OVERVIEW OF TAYLOR SERIES

- **Motivation**
  - Solving Complex Problems:
    - Vital in physics, engineering, and various scientific disciplines.
    - Facilitates precise approximations, aiding in solving complex problems that involve non-linear functions.

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OVERVIEW OF TAYLOR SERIES

- **Motivation**
  - Optimization and Numerical Analysis:
    - Used in optimization algorithms and numerical analysis to find approximate solutions.
    - Essential for iterative methods and understanding convergence properties.

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TAYLOR SERIES IN ONE DIMENSION

- **Taylor Series in One Dimension**
  The Taylor series is a representation of a function as an infinite sum of terms calculated from the values of its derivatives at a single point. In one dimension, the Taylor series for a function $f(x)$ is given by:
  $$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$
  where a is the point around which the series is expanded and $f^{(n)}(a)$ represents the n-th derivative of f evaluated at a.

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TAYLOR SERIES IN ONE DIMENSION

- **Taylor Series Expansion around a Point**
  The Taylor series expansion provides a local approximation of a function around a specific point. The general formula for the Taylor series expansion around a point a is:
  $$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$
  This expansion allows us to approximate the function using a polynomial that captures its behavior near the chosen point. The accuracy of the approximation improves as more terms in the series are considered.

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TAYLOR SERIES IN ONE DIMENSION

- **Understanding Terms**

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

  - Each term represents the contribution of a higher-order derivative.
  - The series captures the local behavior of a function around a point.

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TAYLOR SERIES IN ONE DIMENSION

- **Importance of Derivatives**

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

  - Derivatives provide information about the rate of change.
  - Higher-order derivatives refine the accuracy of the approximation.
  - Taylor Series helps approximate functions beyond simple polynomials.

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TAYLOR SERIES IN ONE DIMENSION

- **Remainder Term:** The remainder term $R_n(x)$ for an nth-degree Taylor polynomial is given by
  $$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$
  where c lies between a and x.
  - The remainder term accounts for the error when using a Taylor polynomial to approximate a function.
  - It quantifies the difference between the actual function value and the value predicted by the Taylor polynomial.

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TAYLOR SERIES IN ONE DIMENSION

- **Estimating Error using Remainder Term:** To estimate the error when approximating $f(x)$ by its nth-degree Taylor polynomial $P_n(x)$ at a specific point x, we use the remainder term:
  $|f(x) - P_n(x)| \leq R_n(x)$
    - The remainder term provides an upper bound on the error of the approximation.
    - Helps in determining how many terms to include in the Taylor series for a desired level of accuracy.

TAYLOR SERIES IN ONE DIMENSION

- **Walkthrough of a simple one-dimensional function**
  - Let's consider the function $f(x) = e^x$.
  - We want to find the Taylor series expansion of $f(x)$ about $x = 0$.
  - The derivatives of $f(x) = e^x$ at $x = 0$ are:
    $f(0) = 1, \quad f'(0) = 1, \quad f''(0) = 1, \quad f'''(0) = 1, \quad \text{etc.}$
  - Therefore, the Taylor series expansion of $f(x) = e^x$ about $x = 0$ is:
    $e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

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TAYLOR SERIES IN TWO DIMENSIONS

The Taylor series expansion for a function of two variables $f(x_1, x_2)$ around the point (a_1, a_2) is given by:

$$f(x_1, x_2) = \sum_{j=0}^{\infty} \left\{ \frac{1}{j!} \left[(x_1 - a_1) \frac{\partial}{\partial x_1'} + (x_2 - a_2) \frac{\partial}{\partial x_2'} \right]^j f(x_1', x_2') \right\}_{x_1'=a_1, x_2'=a}$$

$$= f(a_1, a_2) + \left[(x_1 - a_1) \frac{\partial f}{\partial x_1'} + (x_2 - a_2) \frac{\partial f}{\partial x_2'} \right] + \frac{1}{2!} \left[(x_1 - a_1)^2 \frac{\partial^2 f}{\partial x_1^2} + 2(x_1 - a_1)(x_2 - a_2) \frac{\partial^2 f}{\partial x_1 \partial x_2} + (x_2 - a_2)^2 \frac{\partial^2 f}{\partial x_2^2} \right] + \cdots$$

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MULTIVARIATE TAYLOR SERIES EXPANSION

Equation (1) can be generalized for a real function in n variables as,

$$f(x_1, \cdots, x_n) = \sum_{j=0}^{\infty} \left\{ \frac{1}{j!} \left[\sum_{k=1}^{n} (x_k - a_k) \frac{\partial}{\partial x'_k} \right]^j f(x'_1, \ldots, x'_n) \right\}_{x'_1 = a_1, \ldots, x'_n = a_n} \quad (4)$$

$$f(x_1 + a_1, \cdots, x_n + a_n) = \sum_{j=0}^{\infty} \left\{ \frac{1}{j!} \left[\sum_{k=1}^{n} a_k \frac{\partial}{\partial x'_k} \right]^j f(x'_1, \ldots, x'_n) \right\}_{x'_1 = a_1, \ldots, x'_n = a_n} \quad (5)$$

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CONDITIONS FOR CONVERGENCE

- Convergence of the Taylor series depends on the behavior of the function and its derivatives.
- For a function $f(x)$ and its derivatives, the Taylor series converges if these derivatives do not grow too fast.
- Conditions for convergence involve examining the behavior of the remainder term and ensuring it approaches zero.

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RADIUS OF CONVERGENCE

- The radius of convergence is a crucial concept in the context of power series, including Taylor series.
- It is the distance from the center point (where the series is centered) to the point where the series converges.
- The Taylor series converges within a certain interval defined by the radius of convergence.

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SUMMARY

In essence, the Taylor series stands as a cornerstone in mathematical analysis, providing a versatile approach to representing functions and facilitating precise approximations. Its impact extends across academic disciplines and practical domains, making it an indispensable tool for researchers, scientists, and engineers alike.

LASU-VLAP

Your attendance for this lecture can not be marked at this moment because it was scheduled for Mon., Jan. 13, 2025 8:00 AM (WAT) - Fri., Jan. 17, 2025 6:00 PM (WAT) and you are outside the specified lecture period.

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